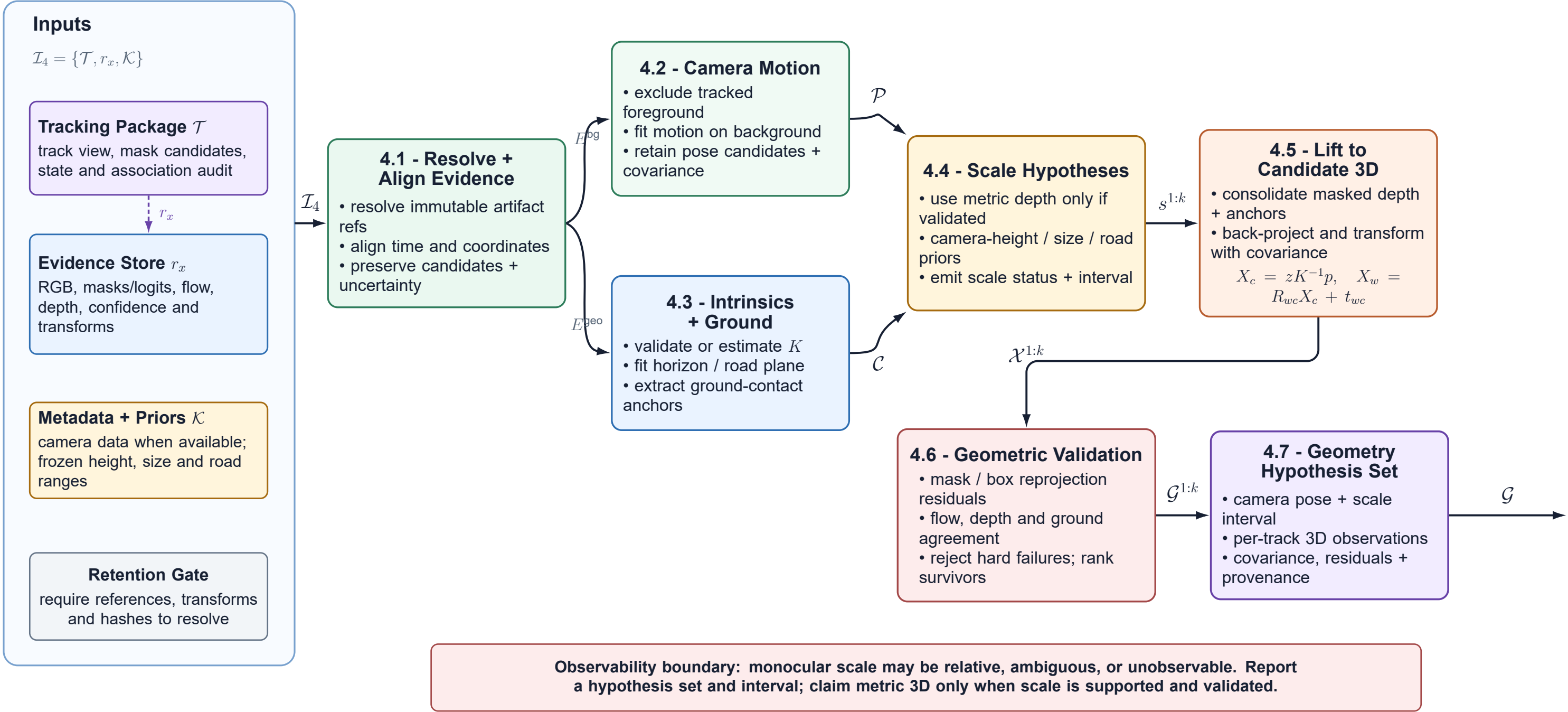


Step 4 - Geometry and Metric-Scale Hypotheses

Output: ranked geometry hypotheses with uncertainty

Transform track-aligned image evidence into validated camera and object geometry



E^{bg} background evidence · E^{geo} geometry evidence · $\mathcal{P}$ pose · $\mathcal{C}$ calibration/ground · s scale · $\mathcal{X}$ 3D candidates · $\mathcal{G}$ geometry

Step 4 reads Step 3 artifacts by reference; it does not rerun YOLO, SAM 2, RAFT or DA3.

Metric motion states and physical trajectories are estimated in Step 5.